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An Improved Random Forest Algorithm (ERFA) Utilizing an Unbalanced and Balanced Dataset to Predict Customer Churn in the Banking Sector

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ABSTRACT Customer churn, the phenomenon of customers discontinuing their relationship with a company, poses significant challenges for businesses in various industries, including the banking sector. In this paper, we address the problem of customer churn in the banking sector by selecting and applying four machine learning algorithms on the dataset of the U.S Bank in the case of Imbalance and Balance to predict the churn of customers from financial institutions such as banks and the reasons that drive them to do so. The algorithms were Logistics Regression, SVM, Neural network and Random Forest. The CRISP model was followed for the necessary processing and analysis, as it became clear after that the Random Forest algorithm provided better results in all cases. Consequently, this paper attempted to provide a better improvement to the results obtained from the default Random-Forest algorithm by understanding its working mechanism and adjusting its hyperparameters. Therefore, a mathematical model was proposed and built for the default Random Forest algorithm as a new version of it called ERFA. The final results of ERFA were positive through an increase in prediction accuracy from 85% to 86% in the case of the imbalanced dataset and from 89% to 91% in the case of balanced dataset. The outcomes of this study can help and motivate bank managers in general and in Yemen in particular to adopt this type of smart strategies to maintain their customers and provide them with a better experience. Furthermore, the mechanism established by this paper may help researchers who are interested in the field of developing smart business models.

INDEX TERMS AI, Banking Sectors, Customer Churn, Machine Learning Algorithms, Random Forest

I. INTRODUCTION

A. BANKING SERVICES AND CHALLENGES

Customers

of numerous banks benefit from several services, including the ability to check account balances, locate ATMs, transfer money, deposit cheques, and much more [1]. Banks handle all of their clients' payments in addition to engaging in local and foreign exchange trading. Furthermore, banks promote economic expansion [2] Banks have an impact on economic development because they are the primary financial **institutions** in every nation [3] .For many years, banks have been the primary responsible party for the majority of financial transactions, cooperating with clients and organizations [3]. Even with all of the benefits and features that banks offer, a surprising number of both developed and developing nations have extremely high customer churn rates [3, 4]

Customers are frequently motivated to leave an organization for both controllable and uncontrollable reasons. [5]. Concerns about quality, price adjustments, privacy, competitive offerings that are alluring, low switching costs, general customer discontent, and a host of other issues are examples of controllable factors. [4, 6] assert that a variety of shifts in the customer's circumstances, such as changes in personal lifestyle, difficulties with family, job relocation, shifts in the economy, international travel, death, and many more, may cause uncontrollable factors.

Yemen claims that a large number of its banks are experiencing a decline in customer loyalty, and some of them have expressed the opinion that the conventional methods they have been employing are insufficient and ineffective. Despite their best efforts, customer adoption, intention, and usage of their services are still in their infancy.

B. CLASS IMBALANCE PROBLEM

In datasets used for data mining, class imbalance has become a prevalent issue. This is a typical issue in the field of

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customer churn prediction. A churned class, which is the most significant class, is labeled in fewer examples than the almost all examples of the not churned class in this kind of problem. Using Under-Sampling, Random Sampling, and Over-Sampling was one solution to the class imbalance issue[3].

C. CUSTOMER CHURN STRATEGY

Three primary strategies were taken into consideration by many financial organizations for the purpose of increasing the competitiveness of businesses: (1) obtaining new customers, (2) boosting sales to current customers, and (3) prolonging the customer retention period [1]. Nevertheless, it was discovered that the third strategy was the most successful when evaluating the significance of these three approaches based on the value of return on investment for each [1]. A U.S. bank dataset has been selected for analysis to develop the research model using data available within the Kaggle classification datasets. The dataset can be accessed via the following URL link: https://www.kaggle.com/datasets/shantanudhakadd/bank-customer-churn-prediction.

D. MACHINE LEARNING ALGORITHMS

Four machine learning algorithms were suggested in this study to address the issue of bank customers quitting. A quick overview of each is given below:

1. LOGISTIC REGRESSION

The most popular algorithm for modeling binary dependent variables is logistic regression (LR). It is a kind of statistical classification model with probability that is primarily applied to classification issues [7]. The method can be applied to a variety of variable combinations and improve the accuracy of customer churn prediction.

2. SUPPORT VECTOR MACHINE

Support Vector Machine (SVM) analyzes data for classification or regression issues and comes with corresponding learning algorithms. The kernel function is the focus of this algorithm. An ideal boundary is established between the potential outputs after the data is transformed using the kernel function[8]. The SVM has been used in a variety of fields, including the insurance and telecommunications industries, although its impact on bank customer attrition has been minimal. Thus, the purpose of this study is to examine the SVM algorithm in the banking industry.

3. NEURAL NETWORK

The structure of a neural network (NN), a nonlinear predictive model that learns via training, is similar to that of a biological neural network. The fundamental units of the network are neurons. The neuron's activation function determines the output. NN was selected for this study due to its characteristics because, as numerous prior studies have shown, it produced excellent results in a large number of the cases in which it was applied.

4. RANDOM FOREST

Random forest (RF) is an ensemble learning technique that uses Decision Trees to solve classification and regression problems. The results are generated using the bagging technique. A collection of weak learners joins forces to create a strong learner through ensemble learning. Bootstrap aggregation, commonly referred to as bagging, was utilized to lower the Decision Tree's variance. To produce accurate results, the predictions from several machine learning algorithms are combined using RF. In this study, RF was used to predict customer churn because it was quick and could handle missing and imbalanced data [5].

II. LITERATURE REVIEW

In today's competitive world, many researchers have tried to study customer churn. Most of the research concentrated on predicting customer attrition through the analysis of customer data using machine learning algorithms. In order to improve customer satisfaction and loyalty, researchers from various nations have undertaken studies to determine the factors that influence customers' intentions to stay with banks and use their services.

[9] research indicated that financial institutions do not have any distinct customer retention strategies in place. According to the study, the majority of banks worldwide concentrate on offering a wide range of goods and services to attract more clients and foster a sense of loyalty among them, which in turn increases client stickiness. While banks offer numerous benefits to their clientele, they are negatively impacted by individual customer attrition.

Numerous industries, including the banking sector, also deal with this issue. For instance, a study by [8, 10], revealed that telecom companies experience voluntary churn. The study found that because churn rate influences both the lifetime customer value and the length of service, it has a significant effect on the customer's lifetime value.

Although contractual churn has been studied more frequently, the researchers [7, 11] have reported that churn commonly arises in transactional dealings that can be either contractual or non-contractual. Consequently, a measure of churn is the churn rate.

According to [9] study, churn rates can be measured at different times in accordance with standard industry practices and vary depending on the company and industry. Furthermore, as mentioned by [7] and [3, 12], churn rates are routinely tracked. Churn management is the process of keeping customers by using preventive measures to counteract customer defection, turnover, or attrition. Customers with the highest-ranking probability scores those with the highest likelihood of quitting the company are the focus of incentive-based retention campaigns.

The study of [13] found that an increasing number of businesses across various industries aim to avoid customer attrition, as they consider customer retention to be crucial for boosting business profits. According [6], customer churn is therefore a crucial part of both customer lifetime value

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(CLV) modeling and customer relationship management (CRM).

A group of researchers conducted that like [5] and [6, 14] customers are often motivated to churn due to both controllable and uncontrollable factors. Controllable factors include quality concerns, price changes, privacy concerns, attractive competitor offerings, low switching costs, general customer dissatisfaction, and many other reasons. Some studies argue that uncontrollable factors may be triggered by various changes in customer circumstances like personal lifestyle changes, family challenges, job relocation, economic changes, death, and many others and this also has been noted in [1, 4, 15].

According to studies by [5] and [6, 14], and others, a variety of controllable and uncontrollable factors frequently encourage customers to leave. Concerns about quality, price adjustments, privacy, competitive offerings that are alluring, low switching costs, general customer discontent, and a host of other issues are examples of controllable factors. According to their studies, a variety of changes in the customer's circumstances, such as changes in personal lifestyle, difficulties with family, job relocation, economic shifts, death, and many others, may cause uncontrollable factors. and this also has been noted in [1, 4, 15].

Additionally, using a variety of data analysis techniques, [5, 7] investigated the impacts of sixty distinct churn predictors (both behavioral and demographic parameters) within gamblers. Analogous research endeavors examining the factors or origins of attrition have been carried out across multiple sectors.

However, Yemen is not an expected country, so what we have mentioned previously reflects Yemen in general because in Yemen there are many banks that deliver many kinds of products and services. According to several meetings with some marketing employees in Yemeni banks like Corporation Agriculture Credit (CAC) Bank, International Bank of Yemen (IBY), Tadhamon Bank (TB), and Yemen Kuwait Bank (YKB), we found that these banks suffering from customer churn towards their services and according to their point of view the traditional strategies they have been using are not sufficient and not that much effective. Despite these efforts, the adoption, intention, and usage of their services by customers is still in its infancy.

A group of researchers, such as [16] [17] [18] [19, 20], focused on using the Random Faust algorithm because of its efficiency in predicting the exit of customers or employees in various sectors, seeking to explain its working mechanism in addition to modifying some of its variables. Most of these researchers built a set of mathematical steps that were somewhat advanced and not simplified, so this paper attempted to present a simple and clear mathematical model that explains its working mechanism. Moreover, this paper modified the default values of the Random Foust algorithm to provide higher accuracy by using a database related to customer behavior (existing customer and departing customer) and in both cases imbalanced and balanced Dataset.

III. STUDY OBJECTIVES

The following three points represented the primary goals of this study:

- This study aims to identify the root causes of customer churn in banking services and develop an intelligent model using popular data mining algorithms. This model can help banking managers devise strategies to enhance customer loyalty in the present and future.
- The study focuses on bridging the gap in the Yemeni banking industry by utilizing four effective algorithms (LR, SVM, RF, and NN). These algorithms will enable a comprehensive understanding of churn factors and customer dissatisfaction with banking services in Yemen, which has been previously unexplored.
- The study also aims to improve the performance of the best model among the four proposed models (LR, RF, SVM, NN) in both imbalanced and balanced datasets. The objective is to develop a mathematical model with enhanced churn prediction accuracy, assisting banks in designing effective customer retention strategies and minimizing customer attrition.

IV. STUDY METHODOLOGY

1. CRISP-DM

The process utilized in this study, known as CRISP-DM (Cross-Industry Standard Process for Data Mining), involves experiment design throughout the research lifecycle. Figure 1 illustrates the CRISP-DM process [21].

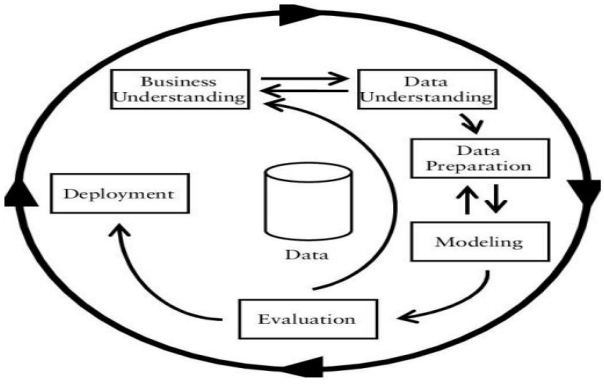

FIGURE 1. CRISP-DM Process (taken from [21])

In this study, the CRISP-DM Process starts with the first stage, which reflects an understanding of the problem represented by the exit of customers from banks. Then the second stage was collecting the necessary data to solve the problem, which was the U.S bank's dataset that consists of

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10,000 bank customer's data records with 18 attributes; 17 attributes as independent factors and 1 as a dependent factor (the target column; Exited)

Then in the third stage, the necessary processing was done for this data by ensuring that it is free of empty values and abnormal values, and not including unnecessary attributes like customer ID, and customer name then solving the problem of categorical attributes and converting it to single numerical values to facilitate dealing with it. moreover, to reduce the range of numerical values to speed up their analysis.
In the third stage, the most significant features (attributes or factors) that contribute to predicting customer churn have been explored as clear in Figure 2.

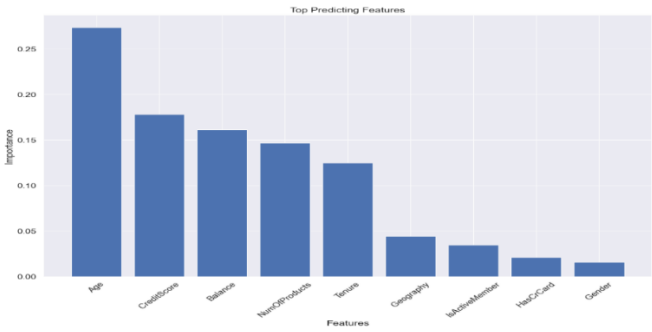

FIGURE 2. The Top Predictive Features in The Dataset of This Study

In the fourth stage, the dividing was made into training data which was 80% of the dataset to build the models, and testing data that were 20 % to ensure the accuracy and quality of the models in prediction.
In the fifth stage, the four algorithms (LR, SVM, NN, RF) were selected in this paper and then used in the process of predicting customer churn.

2. THE ACCURACY MEASURE

The Accuracy measure was made with both datasets that have been used in this study; imbalance and balanced dataset. This is displayed in two graphs to easily differentiate and evaluate the best model.

- **IN IMBALANCE DATASET**

In this study and as it is clear RF with 86% accuracy in the imbalance dataset was the best model, as it is displayed in the histogram in Figure 3.

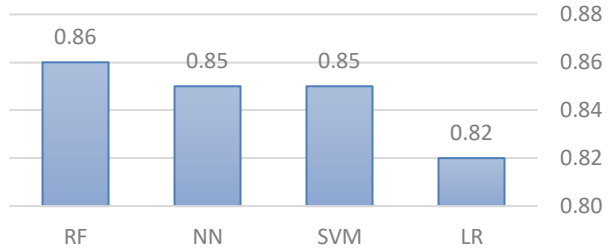

FIGURE 3. The Comparison of The Accuracy of The All Models in This Study

- **IN BALANCE DATASET**

In a balanced dataset, RF has also given the highest accuracy with 89% the same degree has been reached with Neural Network in this case, as it is displayed in the histogram in Figure 4.

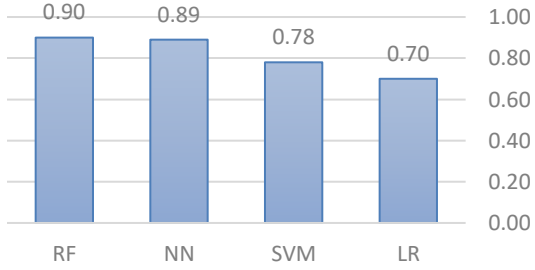

FIGURE 4. The Comparison of The Accuracy of The All Models in This Study

3. RECEIVER OPERATING CHARACTERISTIC (ROC) CURVE

The trade-off between the true positive rate and the false positive rate of LR, SVM, NN, and RF, thresholds is represented visually by the Receiver Operating Characteristic (ROC) curve [22]. Figure 24 exhibits the RF model, which has a higher true positive rate and a lower false positive rate. Its curve is closer to the upper-left corner of the plot, demonstrating better performance.
A higher AUC indicates better discrimination ability for RF in imbalance and imbalance datasets, as Figures 5 and 6 explicitly show. The Area Under the ROC curve (AUC-ROC) is a commonly used summary measure of the model's overall predictive performance.

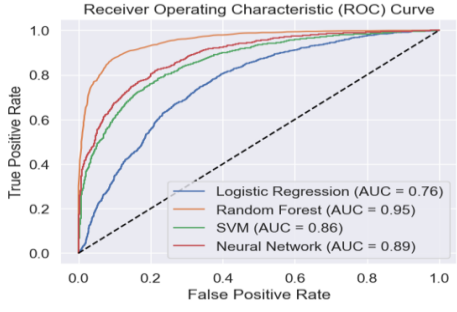

FIGURE 5. The Receiver Operating Characteristic (ROC) In Imbalance Dataset

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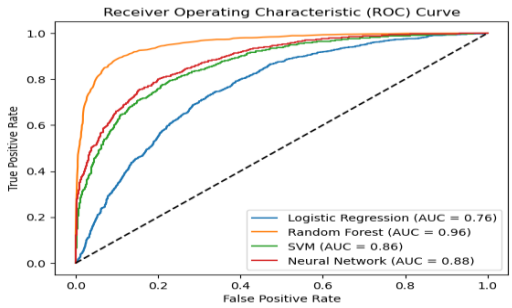

FIGURE 6. The Receiver Operating Characteristic (ROC) In Balance Dataset

4. Models' Evaluation

This study has made a summary of all final results of machine learning models after implementing confusion matrix and classification reports metrics to the four models (LR, SVM, NN, and RF) with balance and imbalance datasets, the final results are displayed in Table.

TABLE 1
THE RESULT OF MACHINE LEARNING MODELS IN THIS STUDY

RESULTS WITH IMBALANCED DATASET															
N	Model	TP	FP	FN	TN	Accuracy	Precision		Recall		F-score		Error	AUC
1	LR	1556	23	324	69	82.30%	0.83	0.62	0.97	0.18	0.9	0.27	17.70%	76
2	SVM	1581	26	266	127	85%	0.86	0.83	0.98	0.32	0.92	0.47	15%	86
3	NN	1543	64	213	180	85%	0.86	0.83	0.98	0.32	0.92	0.47	15%	89
4	**RF**	**1554**	**71**	**199**	**185**	**86%**	**0.89**	**0.72**	**0.96**	**0.48**	**0.92**	**0.58**	**14%**	**95**
RESULTS WITH BALANCED DATASET														
N	Model	TP	FP	FN	TN	Accuracy	Precision		Recall		F-score		Error	AUC
							class 0	class1	class 0	class1	class 0	class1		
1	LR	1156	495	456	1078	70%	0.72	0.69	0.7	0.7	0.71	0.69	30%	76
2	SVM	1296	355	343	1191	78%	0.79	0.77	0.78	0.78	0.79	0.77	22%	86
3	NN	1457	194	160	1374	89%	0.9	0.88	0.88	0.9	0.89	0.89	11%	88
4	**RF**	**1470**	**181**	**145**	**1389**	**90%**	**0.91**	**0.88**	**0.89**	**0.91**	**0.9**	**0.89**	**11%**	**96**

To sum up, as is clear in Table 1 many experiments have been conducted to compare the best result for all algorithms in this study in the case of imbalance and balance. After completing the set of experiments, it became clear that the RF algorithm was the most accurate in both cases using the Python programming language.

V. THE PROPOSED MATHEMATICAL MODEL IN THIS STUDY

As we have noted in the section of the study objective in this paper, one of the most important aims was to improve the accuracy of predicting customer churn in banking sectors. As a result, four famous algorithms; LR, SVM, RF, and NN were implemented. Finally, it was noted that the default RF algorithm provided the best display of accuracy out of all with 86% in the case of the imbalance dataset and 89% in the case of the balance dataset.

This paper also takes the consideration of some related works like [16] [17] and [19] by introducing and presenting an easy mathematical model trying to enhance and improve the prediction of the RF algorithm, through a set of important assumption steps that make it possible to reach better prediction accuracy as follows:

1- Choosing A dataset from the Kaggle website that represents the field of research, called "Custom Chun", in CSV extension, it was then viewed and understood by converting it into an Excel file to make it easy to read.

2- Doing modification preprocessing to the dataset because the primary goal of this study was to reach the best features that will help in predicting and that would give better accuracy of customer behavior in the banking sectors using the four famous machine learning algorithms.

3- Presenting a set of mathematical hypotheses (hyperparameter process) to improve and enhance the RF algorithm further because it was the best in the accuracy from the three other algorithms (LR, SVM, NN) more than this taking considerations of some recommendations and limitations in previous studies in different field that focus on customer churn.

4- Finally, the results were presented, discussed, and compared with previous studies.

A. STEPS CREATION OF ERFA

As it has been known RF has its general process, and here in this research, we will try to enhance its accuracy by moderating it and restructuring its main operation steps through tuning its hyperparameter.

First, we will take a dataset as an input. Then we will take many steps to create the proposed mathematical model which we call the Enhanced Random Forest Algorithm (ERFA). Finally, produces classification prediction as output. The structure of the mathematical model will contain three main steps as fellow:

1. Input:

- Dataset: A labeled dataset named "ChurnModelling.csv" consisting of input features (F); F= (f1, f2, f3, ... f17) and corresponding output labels (Y).
- The dataset that has been selected contains 10,000 customers, thus row r=10,000 and features F=17.

  We assume ChurnModelling.csv=DF and the DF is a matrix with 17 independent features from f1 to f17, and with 1 dependent label Y with rows r =10,000, the equation matrix will be:

  DF= [[f1…f17], [Y]]

  For checking whether Df is a balanced or imbalanced the following equation will be used:

  $$\text{Gini (DF)}=1-\sum_{V\in(i1,i2,in)}^{n}(p_i)^2$$

  Where n is the total classes in the target label starting from (i 1 to i n,) pi is the probability of every class in DF. Thus, the equation in this study will be:

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Gini (DF)=1-$\sum_{v\in(churn,retain)}^{2}(p_i)^2$

Where;
V *takes the two clases of DF target lable Y* $(churn\&retain)$

2. Process

2.1. *Data Preparation:*

- Input the Dataset DF
- Display all DF attributes to understand it.
- Do some preparation of DF (clean noise values, fix null cells, encoded and ranking classification variables).
- For features selection, not all features will be selected only 14 will be used; customer Id, customer name, and complaint features will be deleted. The only rest features selected will be included in the model (algorithm) if the correlation of 1 fs and Y ≥ 10%. This will be done to improve the efficiency of the prediction and to determine the weak correlation.

2.1. Data split:

- Spilt DF to the training set (TS) and Validation set (VS): TS= DF*75% and VS= DF*25%.

2.2. Bootstrap Sampling:

- Create multiple bootstrap samples (many random subsets of the TS).In this study, we will assume we have 30 random subsets with replacement of the TS with 30% out of bagging simple in each tree for TS validation.

2.3. Building Decision Trees:

before building 30 trees we have to determine how many and what features will be selected fs to create every node starting from the root up to reaching nodes that might be before the leaf, this will be determined by this equation:

$$fs = \sqrt{16} = 4$$

The number 16 means that all features will be included in the analysis except the complaint and exited columns in the dataset DF.

Thus, the fs that will be selected are equal to 4 in every node creation. For creating subgroups of all features, we have to calculate every threshold, and this will be done by taking the median of all fs. Then for easy computation, the Gini index will be used to calculate the importance of the 4 random fs that were selected in each of 30 random datasets, and to create these 30-decision trees, we have two steps:

1) Calculate the Gini (weight) for each fs (here there are only 4) that was selected randomly, to do this, the first thing we have to calculate every class belonged to the same fs by the following equation:

$$\text{Gini (fsi, i1:in)} = 1 - \sum_{i=1}^{n}(p_i)^2$$

Where n is the number of classes in all fs and pi is the probability of every class in the same fs (fsi could be fs1 or fs2…. or fsn)

2) Then calculate the whole weight of fs in the 30 new datasets (bootstrapping) and after measuring the highest one, the root will be created then the nodes for the second; we keep doing this until we reach the node before the leaf which has the least weight in all trees, this will be done by this equation:

$$\text{Gini Index (fs)}=1-\sum_{V\in(i1,i2,in)}^{n}\frac{|TSv|}{TS}W(TSv)$$

Where W(TSv) is the weight for the whole TS target label (Y). The split that provides the lowest Gini index value is chosen as the best split for the data. Repeating this process to determine multiple splits in the decision tree contributes to building the tree and defining the conditions used for decision-making.

In this study, for have a deep analysis of customer churn features in all cases, we assume the maximum depth of all trees in the bootstrapping set as follows :

Tree Depth =26

After implementing all these procedures and stages, we will reach multiple trained decision tree models that are ready to deal with any new simple in the VS or any external values not included in DF.

To sum up, these previous hyperparameters will create a new version of RF and in this study, we called it ERFA, this will involve a loop or iteration over the 30 trees. Each iteration includes bootstrap sampling, feature selection, and decision tree construction.

In the coming sections, we will ensure that this modification of RF; the new model ERFA, will give better results than the original RF and also better in the accuracy of one of the modern related studies in this field.

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3. **Output**

Adding each trained tree to the ensemble. The final prediction of classification is obtained by aggregating the outputs of all the trees in the ensemble. The Prediction =The most frequent class (churn/ or retain) in all 30 trees. In other meaning every decision tree will give us the result of its scenario or path by using the majority (most frequent class from all trees). This process is often called voting. The prediction can be made by inputting any simple from the VS.

Therefore, all-adjusted decision trees that have been made are only to create the ERFA and the final output of ERFA is a set of classifications for new (unseen data).

B. EVALUATION

For ensuring and evaluating the quality of ERFA in prediction, some important performance metrics will be applied like Confusion Matrix contents (TP, FP, FP, TN), and classification report elements as follows:

1- Accuracy $=\frac{TP+TN}{TP+TN+FP+FN}$

2- Recall $=\frac{TP}{TP+FN}$

3- Precision $=\frac{TP}{TP+FP}$

4- Specificity $=\frac{TN}{FP+TN}$

5- F1-score(measure) $=2\times\frac{\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}$

Generally, the accuracy will be calculated from the ERFA decision and this will be by all 30 tree outputs. It will be measured by the majority of the two classes count, through these two conditional equations:

1) If Churn > Retain then the predicted class will be churn
2) If Churn < Retain then the predicted class will be Retain

6- AUC ROC & AUC PR drawing

Both AUC ROC and AUC PR may allow comparison of the quality of the mathematical proposed model "ERFA" that is proposed in this study with original RF (default parameters of RF) in the two cases: imbalance and balance dataset.

3. THE RESULT OF IMPLEMENTATION THE MATHEMATICAL MODEL (ERMA)

Two experiments have been done to see how different could be in the results of the two datasets that we have; imbalanced and balanced datasets.

A. EXECUTE ERFA WITH IMBALANCE DATASET

Like any normal algorithm that is implemented. First, the necessary libraries that will be used must be called. Then, the dataset has to be fetched and checked surely if it is free of problems such as empty values, duplicate values, abnormal values, etc. After that, the encoding should be done for the categorical values. Then, the large values must be made between -2 and + 2, to make it easier for the algorithm to deal with it. In addition, the important dependent variables that influence the churn of customers must be shown, so that they can be selected in the algorithm and to facilitate the learning process for the algorithm.

According to the mathematical model proposed in this study, many default variable values have been changed to new values, which could help in raising the accuracy rate of the random forest algorithm. Some modifications were made based on the mathematical model proposed in this study.

At first, the default division of training data and test data was modified to allow the random forest model to learn with more data. The data was divided into 25% as test data and 75% as training data and the random state was set to 40. Figure 7 shows the modified split of the dataset.

```
from sklearn.model_selection import train_test_split

df_aftersplit=df_copy.copy()
# Assuming 'df' is the DataFrame you want to split
X = df_aftersplit.drop('Exited', axis=1)  # Features (remove the target variable column)
y = df_aftersplit['Exited']  # Target variable

# Split the data into train and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_state=40)

# Print the shapes of the resulting train and test sets
print("Train set shape:", X_train.shape, y_train.shape)
print("Test set shape:", X_test.shape, y_test.shape)
```

FIGURE 7. The Modified Split of The Dataset

After that, the three basic values in the default RF algorithm in Python were modified; the number of trees (n_estimators = 30), the criterion = 'gini', the number of variables chosen when creating the tree = 4, and the max depth=26. Figure 8 shows the code with adjusted values.

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```
from sklearn.ensemble import RandomForestClassifier


# Create a Random Forest model
rf = RandomForestClassifier(n_estimators=30,criterion='gini',max_features=4,max_depth=26)
#rf = RandomForestClassifier()

# Train the model on the training data
rf.fit(X_train, y_train)

# Make predictions on the test data
y_pred = rf.predict(X_test)

# Evaluate the model performance
accuracy = rf.score(X_test, y_test)

# Print the accuracy
print("Accuracy:", accuracy)
from sklearn.metrics import classification_report

# Print classification report
print(classification_report(y_test, y_pred))

cm = confusion_matrix(y_test, y_pred)

# Print the confusion matrix
print(cm)
```

FIGURE 8. The Code with Adjusted Values

After running the above code, it can be noted that the accuracy of the overall ERFM has been improved and reached 87 % in the classification report with also better results of confusion matrix compared with the default RF 86% that have been shown in Table 1. Figure 9 shows the classification report values and the confusion matrix of ERFM.

```
              precision    recall  f1-score   support

         0.0       0.89      0.96      0.92      2011
         1.0       0.75      0.50      0.60       489

    accuracy                           0.87      2500
   macro avg       0.82      0.73      0.76      2500
weighted avg       0.86      0.87      0.86      2500

[[1931   80]
 [ 245  244]]
```

FIGURE 9. The Classification Report and The Confusion Matrix Values Of ERFM

This study also presented an illustration of the ROC curve that shows the extent of the prediction accuracy of ERFM, as it reached 95% in the case of an imbalance dataset, which is a value close to one. Figure 10 shows the ROC curve in the case of an imbalance dataset.

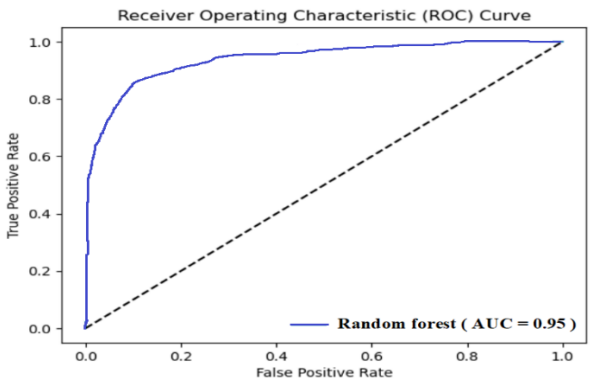

FIGURE 10. The ROC Curve in the Case of An Imbalance Dataset

B. EXECUTE ERFA WITH BALANCE DATASET

The same procedures that were done to the imbalanced dataset were done to the balanced dataset. The only main difference between them is in the balanced dataset the number of instances (customers) was more than the instances in the imbalance dataset. The instances in imbalance were 10,000 customers but the instances in the balance dataset have been increased to reach 15,924 customers. This experiment was done to overcome the problem of the biased learning of the algorithm with the minor class (class 1) by running the code of resample in SMOTE (). Figure 11 shows the code for resampling the original dataset.

```
import pandas as pd
from imblearn.over_sampling import SMOTE

# Initialize SMOTE
smote = SMOTE()

# Resample the data using SMOTE
X_resampled, y_resampled = smote.fit_resample(X, y)
df_resampled = pd.concat([X_resampled, y_resampled], axis=1)

ax = sns.countplot(x= df_resampled ["Exited"])
ax.set(title='Number of exited customers after balancing the dataset')
ax.set_xticklabels(['No','Yes'])
plt.show()
df_resampled.head()
df_resampled.shape
```

FIGURE 11. The Code of Resampling the Original Dataset

Also, the spilt of the balanced dataset has been made as Figure 12 display

```
from sklearn.model_selection import train_test_split

df_aftersplit=df_copy.copy()
# Assuming 'df' is the DataFrame you want to split
X = df_resampled.drop('Exited', axis=1)  # Features (remove the target variable column)
y = df_resampled['Exited']  # Target variable

# Split the data into train and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_state=40)

# Print the shapes of the resulting train and test sets
print("Train set shape:", X_train.shape, y_train.shape)
print("Test set shape:", X_test.shape, y_test.shape)
```

FIGURE 12. The Modified Split of The Balanced Dataset

After that, the code for implementing the RF was written with the proposed hyperparameters adjusting to create the ERFA with a balanced dataset. Figure 13 shows the code ERFA with a balanced dataset.

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```
from sklearn.ensemble import RandomForestClassifier


# Create a Random Forest model
rf = RandomForestClassifier(n_estimators=30,criterion='gini',max_features=4,max_depth=26)
#rf = RandomForestClassifier()

# Train the model on the training data
rf.fit(X_train, y_train)

# Make predictions on the test data
y_pred = rf.predict(X_test)

# Evaluate the model performance
accuracy = rf.score(X_test, y_test)

# Print the accuracy
print("Accuracy:", accuracy)
from sklearn.metrics import classification_report

# Print classification report
print(classification_report(y_test, y_pred))

cm = confusion_matrix(y_test, y_pred)

# Print the confusion matrix
print(cm)
```

FIGURE 13. **The Code of ERFA With the Balanced Dataset**

After running the above code, it can be clear that the result of ERFA in the balanced dataset has been improved more and reached 91% accuracy of the overall prediction in the two classes (class 0 and class 1) with also enhanced confusion matrix values. Figure 14 shows the classification report and the confusion matrix values of ERFM.

```
              precision    recall  f1-score   support

         0.0       0.90      0.91      0.91      1955
         1.0       0.91      0.91      0.91      2026

    accuracy                           0.91      3981
   macro avg       0.91      0.91      0.91      3981
weighted avg       0.91      0.91      0.91      3981

[[1780  175]
 [ 190 1836]]
```

FIGURE 14. **The Classification Report and The Confusion Matrix Values Of ERFM**

With the same mechanism, an illustration of the ROC curve was also made showing the extent of the prediction accuracy of the proposed model, as it reached 96% in the case of a balance dataset, which is a value greater than 95% in the case of an imbalance dataset. Figure 15 shows the ROC curve in the case of balanced dataset.

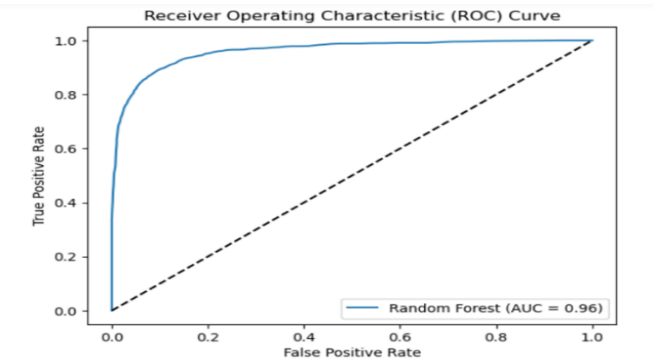

FIGURE 15. **The ROC Curve in The Case of Balanced Dataset**

4. DISCUSS THE RESULTS OF ERFA IN THE TWO EXPERIMENTS

It was clear from the two previous experiments (in the case of the imbalanced and balanced dataset) that the results of the mathematical model of the RF algorithm that was proposed and explained which we called in this study "ERFA" are more accurate and efficient than the default RF algorithm that was explained and shown in Table 1. Moreover, the results obtained from ERFA were more efficient (fast in the running; only 30 decision trees were used) than many previous studies, as an example of this is the Internet Banking study of [23] whose results reached only 81% of accuracy with much more decision trees that reached to 50.

It was confirmed that the result was improved in predicting customer churn with adjusted hyperparameters of RF algorithm which we called ERFA that have been explained and created mathematically in detail and then implemented programmatically on Python.

5. THE CONCLUSION

This paper addresses the problem of customer churn in the banking sector. This study proposed four machine learning algorithms namely Logistic Regression (LR), Support Vector Machine (SVM), Neural Network (NN), and Random Forest (RF) for examining their efficiency in the process of predicting customer churn. Whereas, the RF algorithm was the best regarding accuracy, precision, recall, F1 score, and area under the receiver operating characteristic curve (AUC-ROC). This study then developed a mathematical model for further understanding RF and issued a new version of it that was named Enhanced Random Forest Algorithm (ERFA), which provided better results than the default Random Forest algorithm results by adjusting some variables related to its implementation using a real dataset, which was the U.S bank customers and with imbalance and balance scenarios.

Through comprehensive experiments, we demonstrate the superiority of the ERFA in churn prediction accuracy. The findings of this study will provide valuable insights to banking institutions seeking to reduce customer churn and improve overall customer relationship management. Banks can proactively identify potential churners, tailor personalized retention strategies, and allocate resources efficiently to maximize customer satisfaction and loyalty.

6. THE CONTRIBUTION OF THIS STUDY

Our proposed mathematical model which builds upon the foundation of RF introduces several enhancements to tackle the limitations of the default approach. Very few researchers have contributed to building procedures trying to enhance their results of predicting customer churn in different sectors by using this kind of strategy. Thus, this study approach may also provide valuable insights to the researchers who are interested in creating new models. In addition, the outcome of this research may motivate various financial institutions to use artificial intelligence methodologies to analyze customer behavior and provide appropriate services to them

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which in the end will lead to maintaining them.

7. THE RECOMMENDATIONS OF THIS STUDY

- Preventing manageable attrition is crucial for retaining customers and offers greater benefits than investing in expensive new client acquisitions.
- Banks can implement targeted customer retention strategies and provide exclusive incentives to individuals who exhibit churning behavior.
- Ideally, banks ought to take a proactive approach and detect possible customers who may be at risk of leaving.
- By increasing the retention rate by as much as 5%, the bank might increase its profits by as much as 85%.
- Yemeni banks should use machine learning algorithms to reduce customer attrition.

8. THE LIMITATION OF THIS STUDY

The U.S. bank dataset was chosen as a simple for the purpose of studying the situation of customer churn in banks generally. An actual dataset was not used to reflect the current situation of customers leaving Yemeni banks due to the difficulty in the data collection process because of security issues and the data confidentiality policy of some and because of the novelty of this type of scientific research based on studying customer behavior and analyzing their data using artificial intelligence tools in Yemen.

One more limitation of this paper is that it only created a mathematical model for the RF algorithm because it gave better results than other algorithms. Also, only four variables were modified in the Random Forest algorithm that has been shown in the mathematical model in this study, so in future research, it is possible to modify the rest of the variables related to RF to obtain results that may be better than those presented in this study.

9. FUTURE WORK

In future work, we look forward for applying the four proposed algorithms that were chosen in this study specially ERFA to predict the churn of customers in different Yemeni financial institutions, especially banks that are working in Yemen because of their major role in raising the total economy of the country.

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